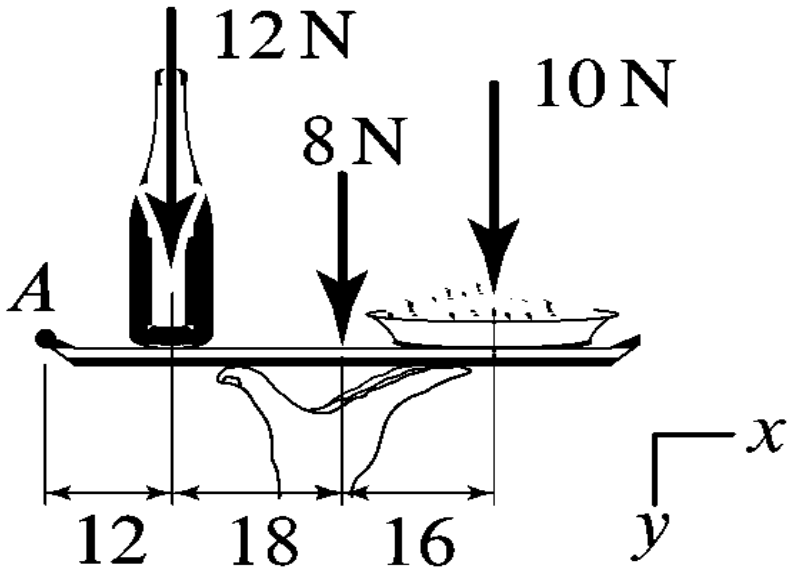
Centroids and Distributed Force Systems

Department of Mechanical Engineering, KNUST
D. A. Quansah, PhD

Centroid and Centre of Gravity

- Every atom of a material has a weight due to gravity, and in aggregate, these yield the total weight.
- For many analysis and design purposes, it is sufficient to know only the total weight of the body.
- This constitutes an equivalent force system.
- The remaining information needed is the location of the line of action for the weight – the center of gravity.

Centroid and Centre of Gravity

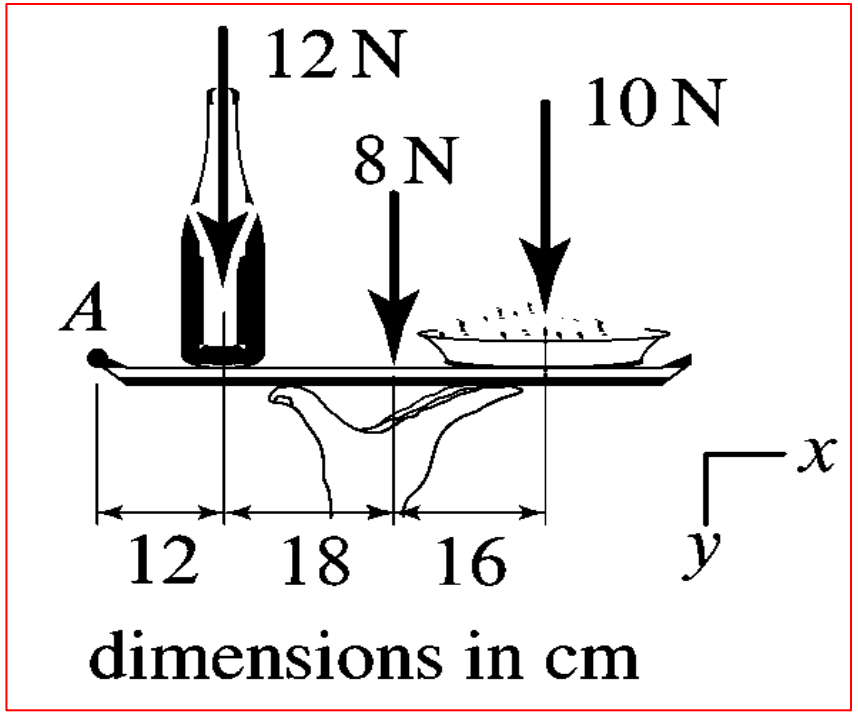


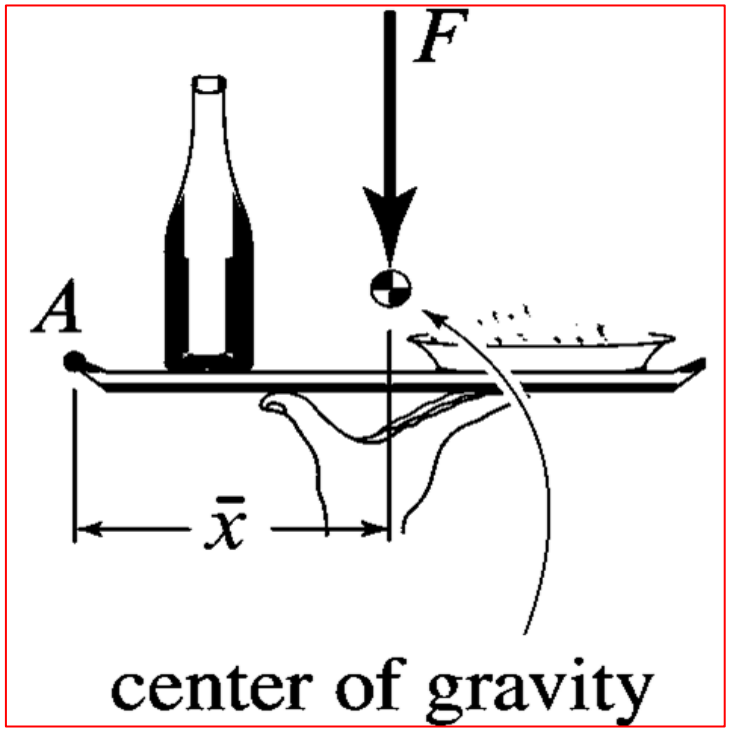
The Figure shows a waiter at a restaurant serving a drink and rice meal on a tray.

The center of gravity is defined to be the average position of the weight distribution of the objects the waiter carries.

The weights of the wine, tray, and rice are 12, 8, and 10 N, respectively. The positions are also shown.

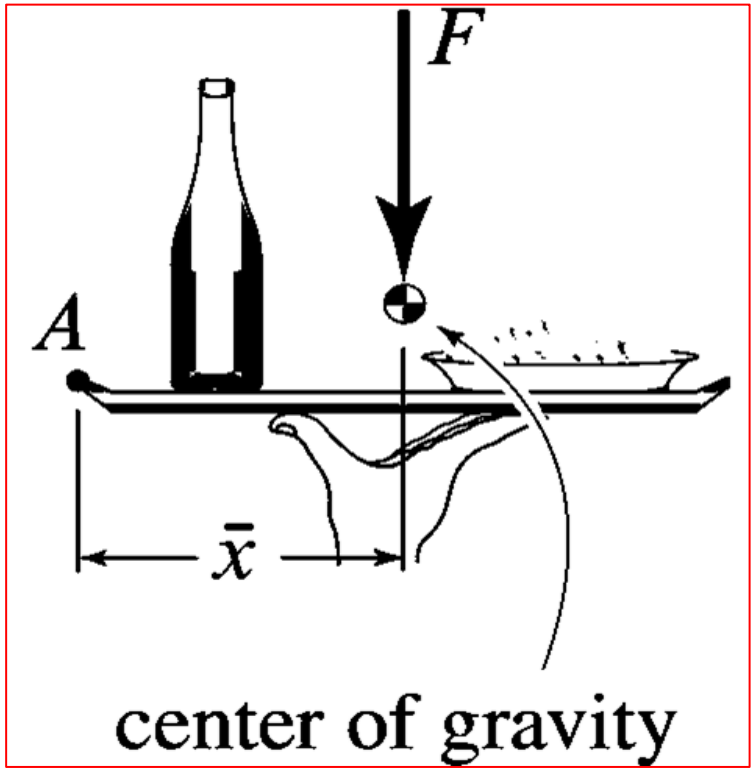
Centroid and Centre of Gravity





The weights could be replaced with a single force, F, which is to be located at the center of gravity for the collection of objects.

Centroid and Centre of Gravity



center of gravity

The weights could be replaced with a **single force, F**, which is to be located at the **center of gravity** for the collection of objects.

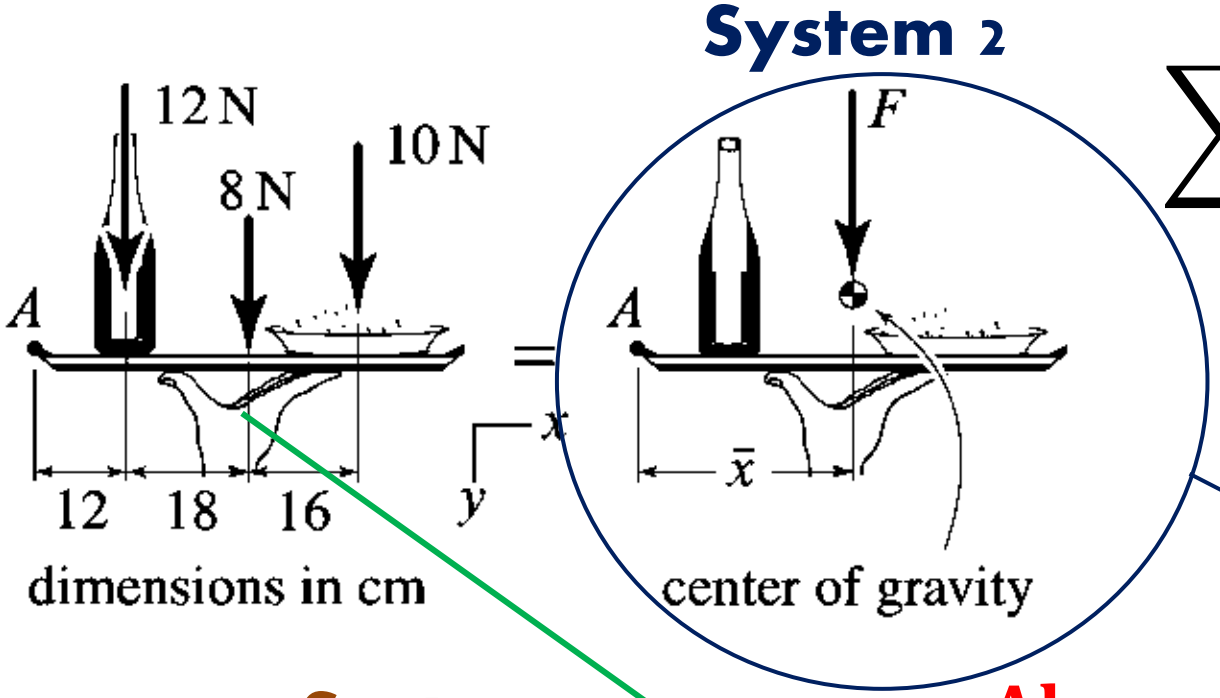
The **force** and its **position** (the **center of gravity**) can be found by constructing an **equivalent force system**.

$$\sum (F_y)_{System1} = \sum (F_y)_{System2}$$

and

$$\sum (M_A)_{System1} = \sum (M_A)_{System2}$$

Centroid and Centre of Gravity



$$\sum (F_y)_{System1} = \sum (F_y)_{System2}$$

$$12\ N + 8\ N + 10\ N = F$$

$$F = 30\ N$$

System 1

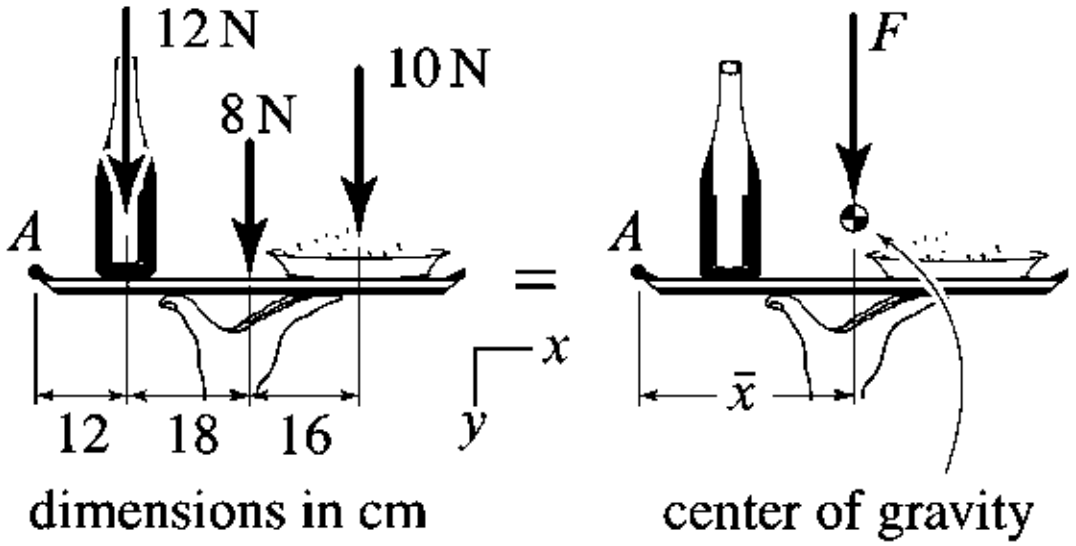
Also:

$$\sum (M_A)_{System1} = \sum (M_A)_{System2}$$

$$(12\ N)(12\ cm) + (8\ N)(30\ cm) + (10\ N)(46\ cm) = F\bar{x} = (30\ N)\bar{x}$$

$$(30\ N)\bar{x} = 844\ N.cm \quad \Rightarrow \quad \bar{x} = 28.1\ cm$$

A generalization may be written as:



$$(12\ \text{N})(12\ \text{cm}) + (8\ \text{N})(30\ \text{cm}) + (10\ \text{N})(46\ \text{cm}) = (30\ \text{N})\bar{x}$$

$$\bar{x} = \frac{(12\ \text{N})(12\ \text{cm}) + (8\ \text{N})(30\ \text{cm}) + (10\ \text{N})(46\ \text{cm})}{30\ \text{N}}$$

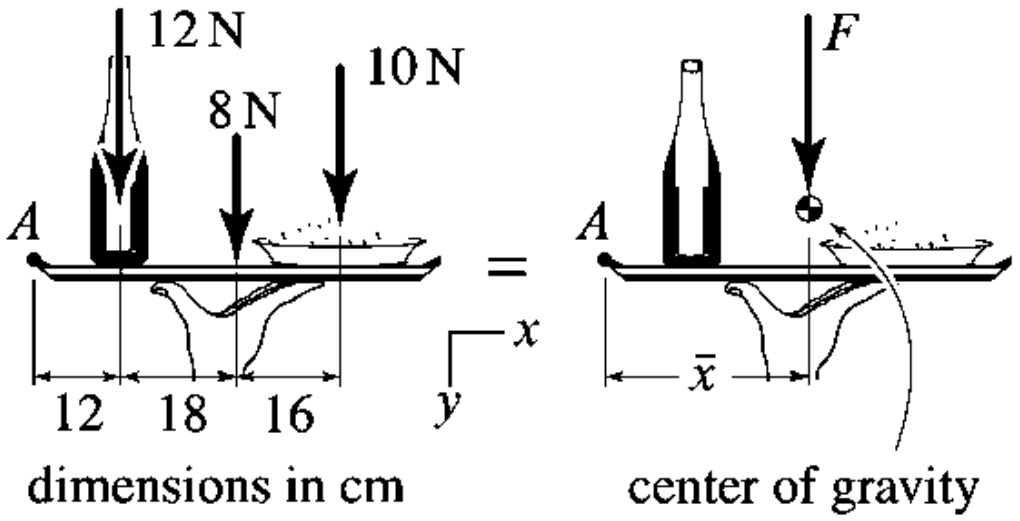
$$\bar{x} = \frac{\tilde{x}_1 W_1 + \tilde{x}_2 W_2 + \tilde{x}_3 W_3}{W_1 + W_2 + W_3}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i W_i}{\sum_{i=1}^{n} W_i}$$

Where:

- n **is the number of objects,**
- W_i **is the weight of object** i**,**
- $\tilde{x}_i$ **is the location of** W_i **(i.e., the moment arm)**
- $\tilde{x}_i$ **is measured from the origin of the coordinate system.**

Centroid and Centre of Gravity



$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i W_i}{\sum_{i=1}^{n} W_i}$$

$\tilde{x}_i W_i$ is the **first moment** of the **weights**

The equation above may be generalized for any distribution – to find its average position.

Replacing the weight, W_i, with mass in the equation yields the average position of mass for a collection of objects – centre of mass.

Centroid and Centre of Gravity

$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i W_i}{\sum_{i=1}^{n} W_i}$$

Similarly, by replacing W_i with the volume V_i of each object, $\bar{x}$ then measures the average position for the volume of a collection of objects - this is called the centroid.

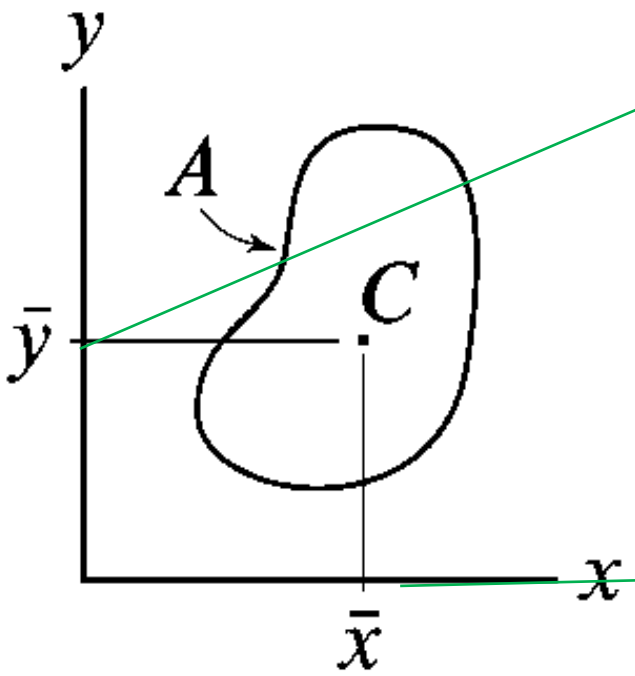
The centroid is defined to be the average position of a distribution of shapes.

The shape can be a line (straight or curved), an area, or a volume.

Formal Definitions

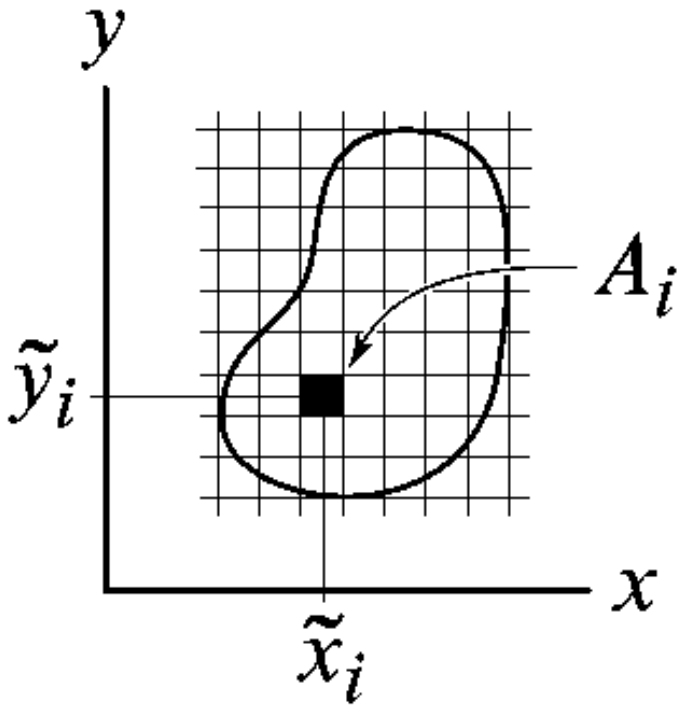
- The **centroid** is the average position of a **distribution of shapes**.
- The **center of mass** is the average position of a **distribution of mass**.
- The **center of gravity** is the average position of a **distribution of weight**.

Centroid of an area



$$\bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i A_i}{\sum_{i=1}^{n} A_i}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i A_i}{\sum_{i=1}^{n} A_i}$$

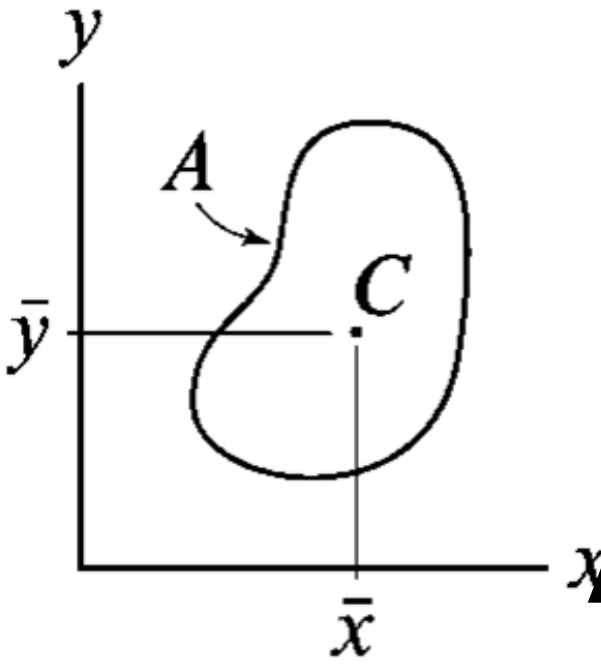


The centroid C of area A is located at $\bar{x}$ and $\bar{y}$.

Area A can be considered as a composite of areas A_i, where the centroid of each of these is located at $\tilde{x}_i$ and $\tilde{y}_i$.

Centroid of an area

$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i A_i}{\sum_{i=1}^{n} A_i} \qquad \bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i A_i}{\sum_{i=1}^{n} A_i}$$



$\sum_{i=1}^{n} \tilde{x}_i A_i$ **is the first moment of Area about the y-axis**

$\sum_{i=1}^{n} \tilde{y}_i A_i$ **is the first moment of Area about the x-axis**

$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i A_i}{\sum_{i=1}^{n} A_i} \qquad \bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i A_i}{\sum_{i=1}^{n} A_i}$$

As $A_i \to 0$

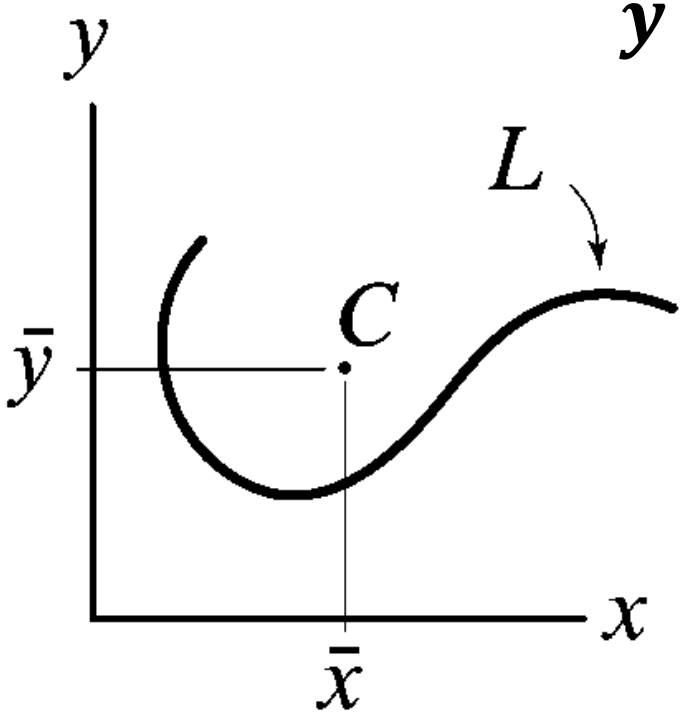
$$\bar{x} = \frac{\int \tilde{x}\, dA}{\int dA} \qquad \bar{y} = \frac{\int \tilde{y}\, dA}{\int dA}$$

Centroid of a line

$$\bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i L_i}{\sum_{i=1}^{n} L_i}$$

As $L_i \to 0$

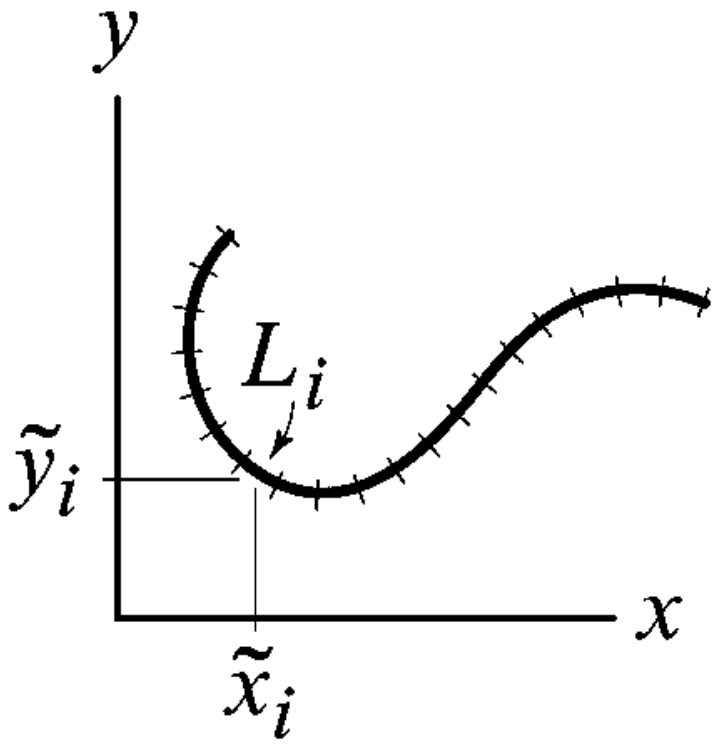
$$\bar{y} = \frac{\int \tilde{y}\, dL}{\int dL}$$



$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i L_i}{\sum_{i=1}^{n} L_i}$$

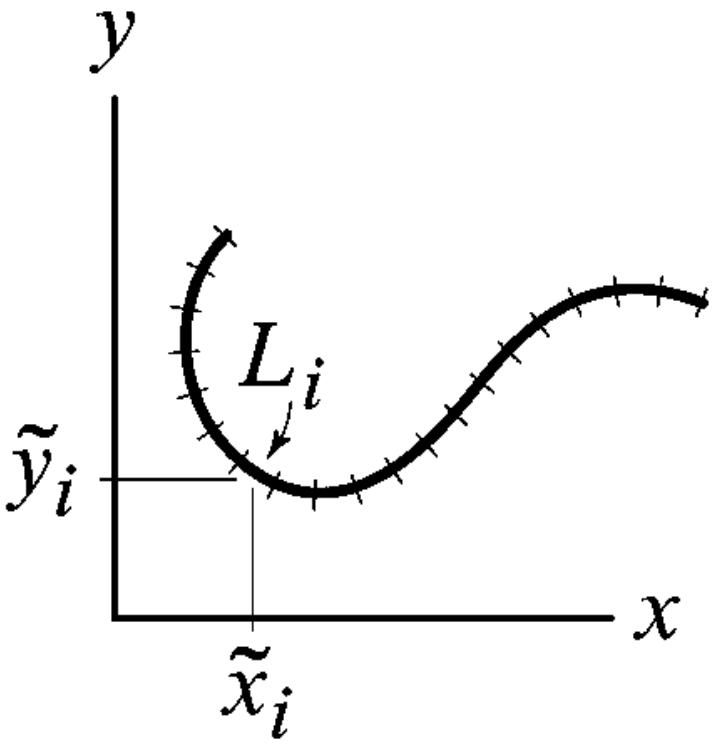
As $L_i \to 0$

$$\bar{x} = \frac{\int \tilde{x}\, dL}{\int dL}$$

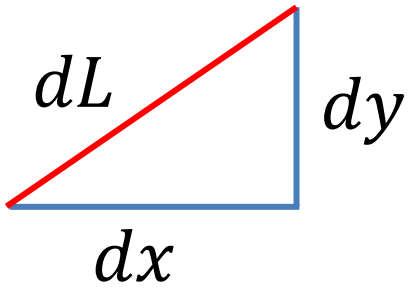


Centroid of a line

$$\bar{y} = \frac{\int \tilde{y}\, dL}{\int dL} \qquad \bar{x} = \frac{\int \tilde{x}\, dL}{\int dL}$$

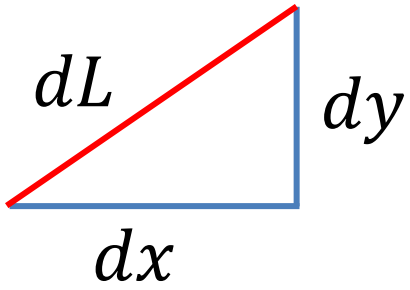


- **It is often more convenient to integrate with respect to x or y.**
- **A transformation between dL and dx or dy is required.**



$$dL = \sqrt{dx^2 + dy^2}$$

Centroid of a line



$$dL = \sqrt{dx^2 + dy^2}$$

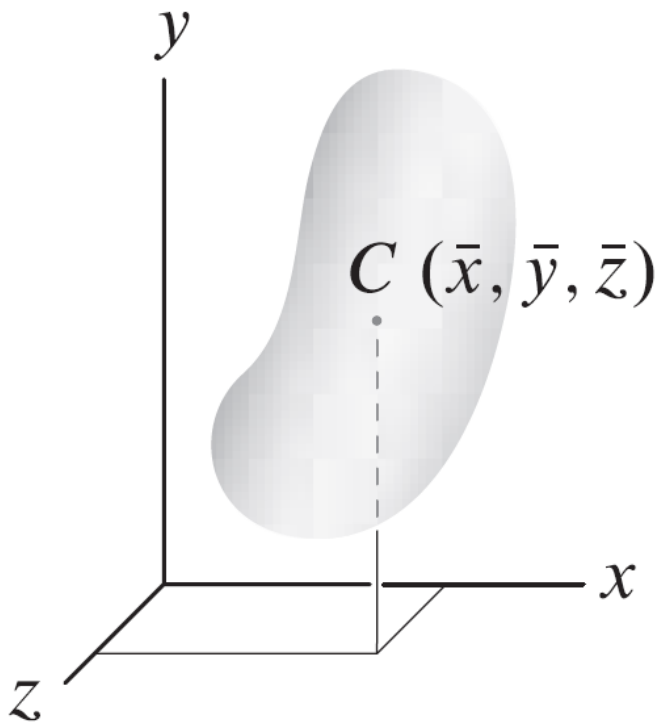
Multiplying right-hand-side by $\frac{dx}{dx}$ and bringing $\frac{1}{dx}$ into the radical yields:

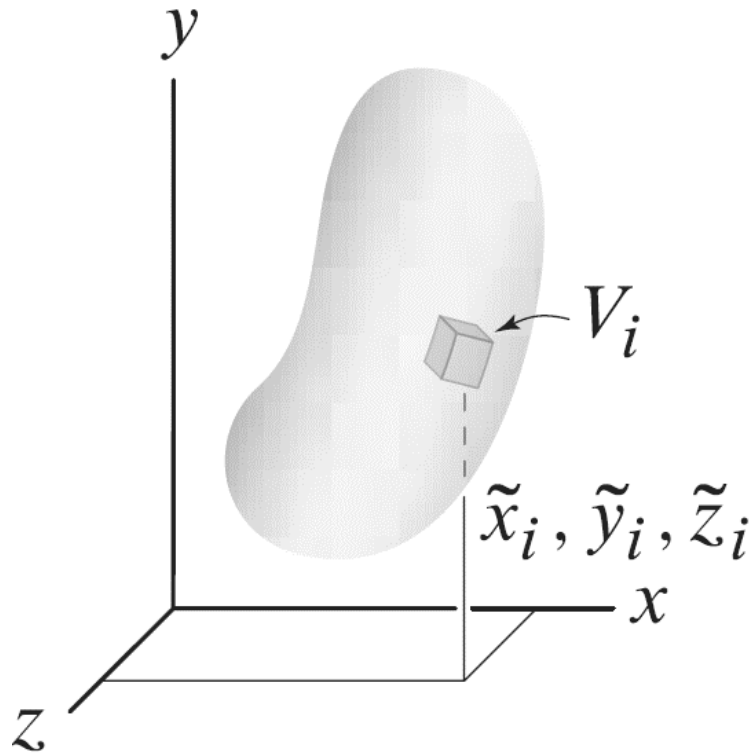
$$dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Similarly, multiplying the right-hand-side by $\frac{dy}{dy}$ yields:

$$dL = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy$$

Centroid of a Volume

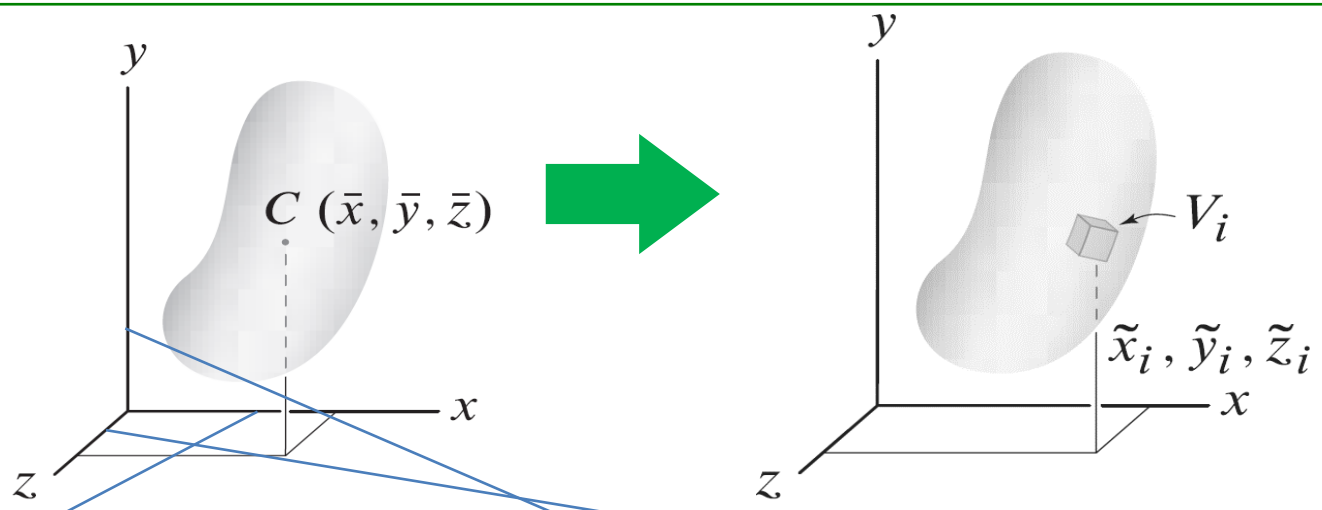




The position of the **centroid C** of the entire volume is $\bar{x}$, $\bar{y}$, and $\bar{z}$.

The volume V may be viewed as a composite, comprising collection of volumes V_i , where the **centroid** of each of these is located at $\tilde{x}$, $\tilde{y}$, $\tilde{z}$.

Centroid of a Volume

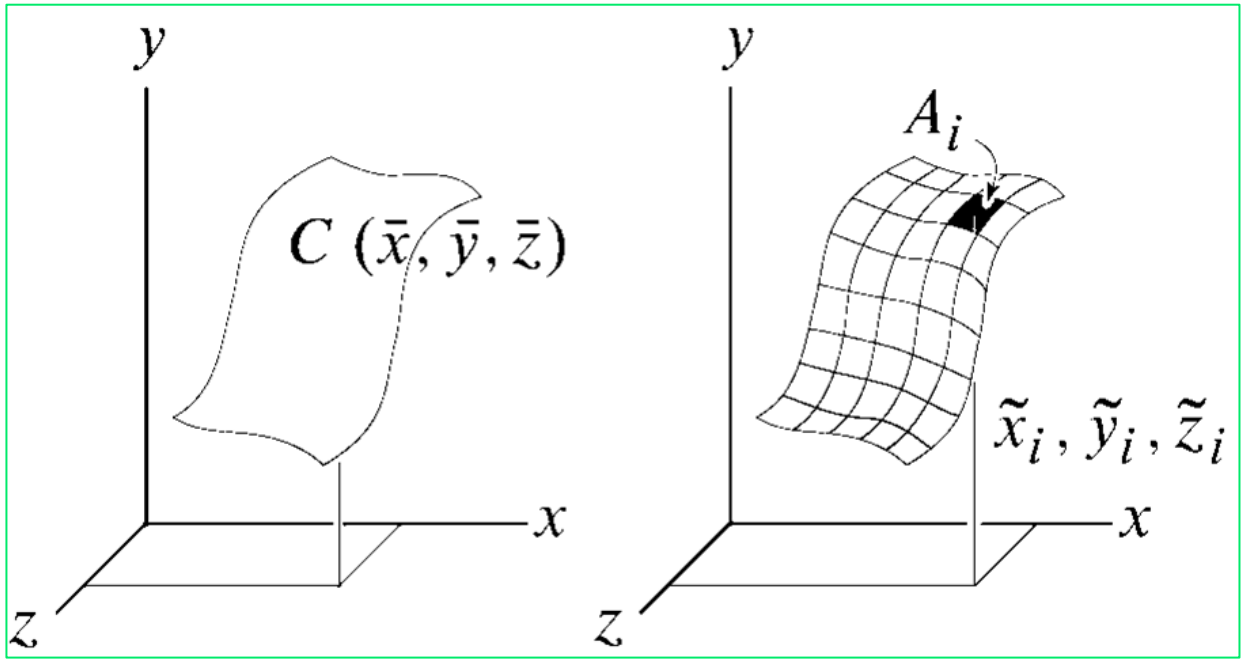


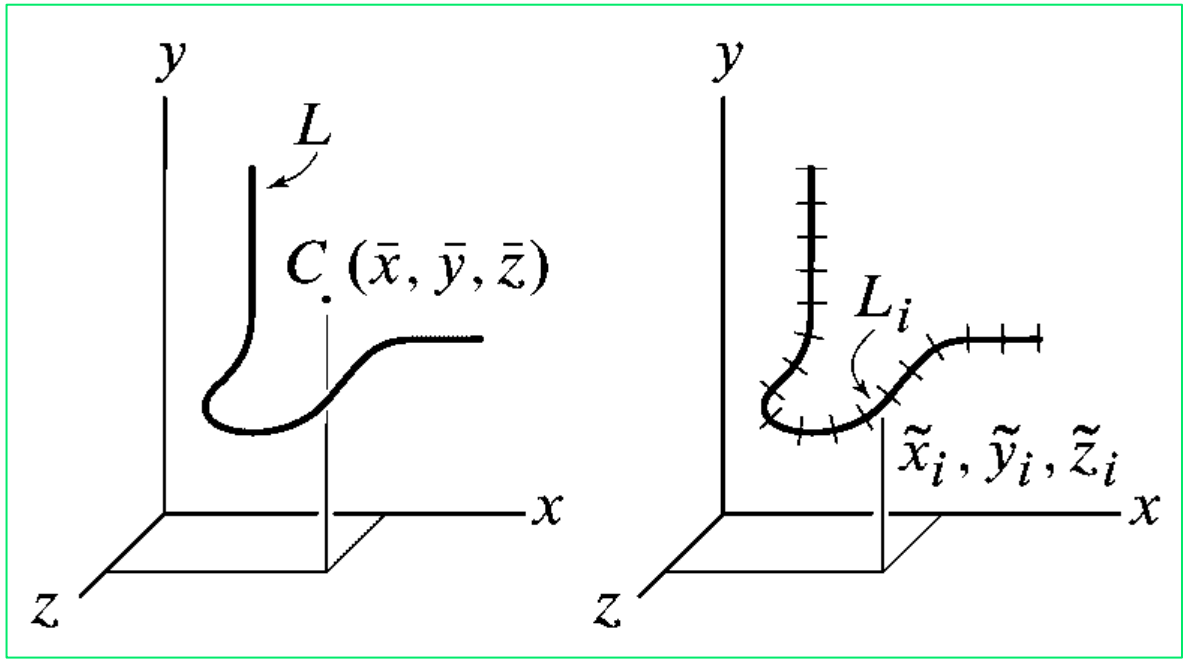
$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i V_i}{\sum_{i=1}^{n} V_i} \qquad \bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i V_i}{\sum_{i=1}^{n} V_i} \qquad \bar{z} = \frac{\sum_{i=1}^{n} \tilde{z}_i V_i}{\sum_{i=1}^{n} V_i}$$

As $V_i \to 0$

$$\bar{x} = \frac{\int \tilde{x}\, dV}{\int dV} \qquad \bar{y} = \frac{\int \tilde{y}\, dV}{\int dV} \qquad \bar{z} = \frac{\int \tilde{z}\, dV}{\int dV}$$

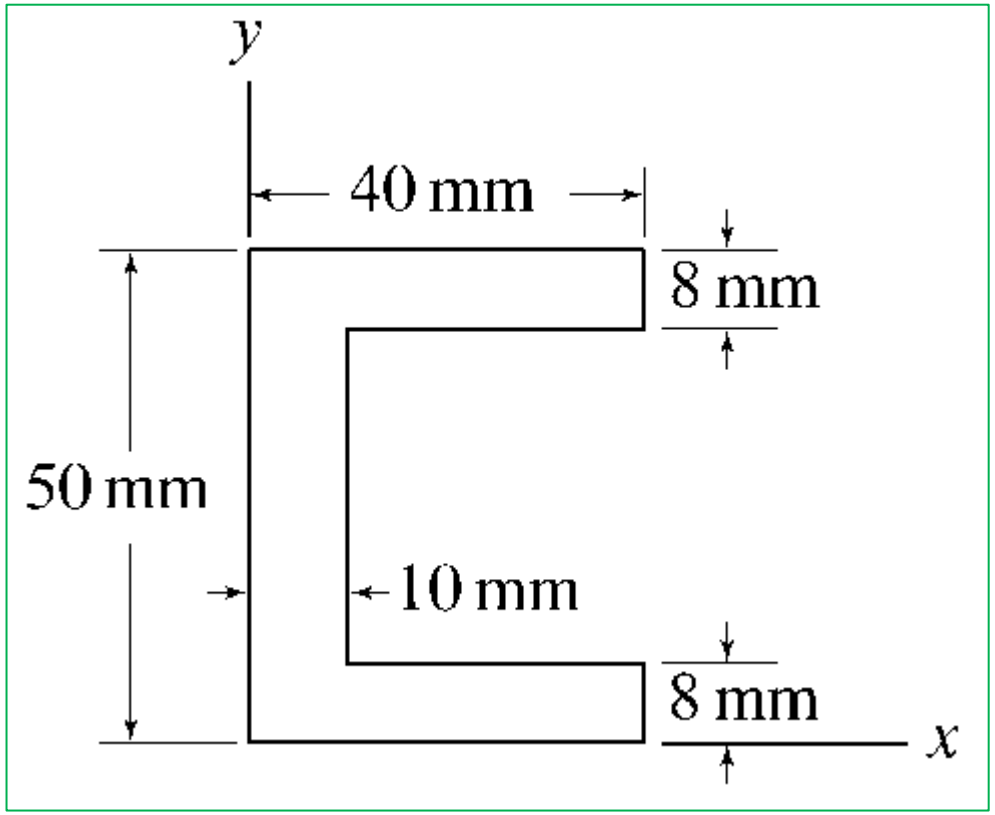
Surfaces and lines in 3-D





1-D and 2-D objects could be located within a 3-dimensional space.

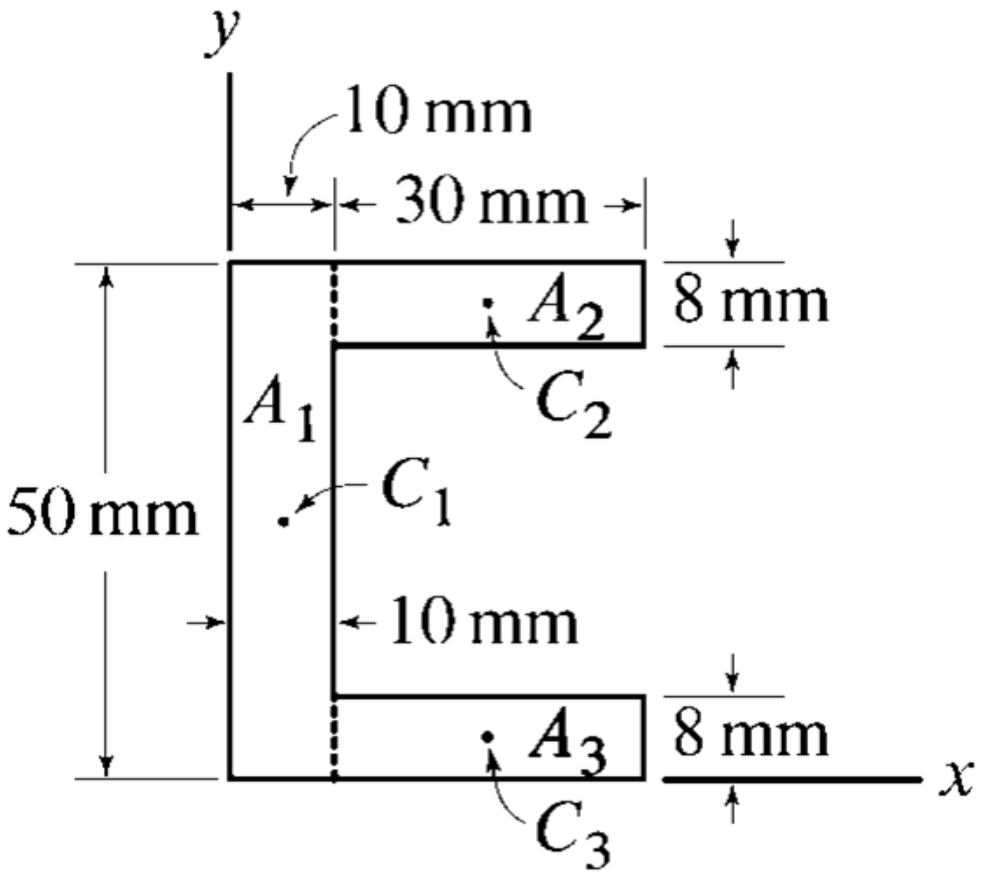
Example



The cross section of an extruded aluminum channel is shown.

Determine the x and y positions of the cross section's centroid.

Example



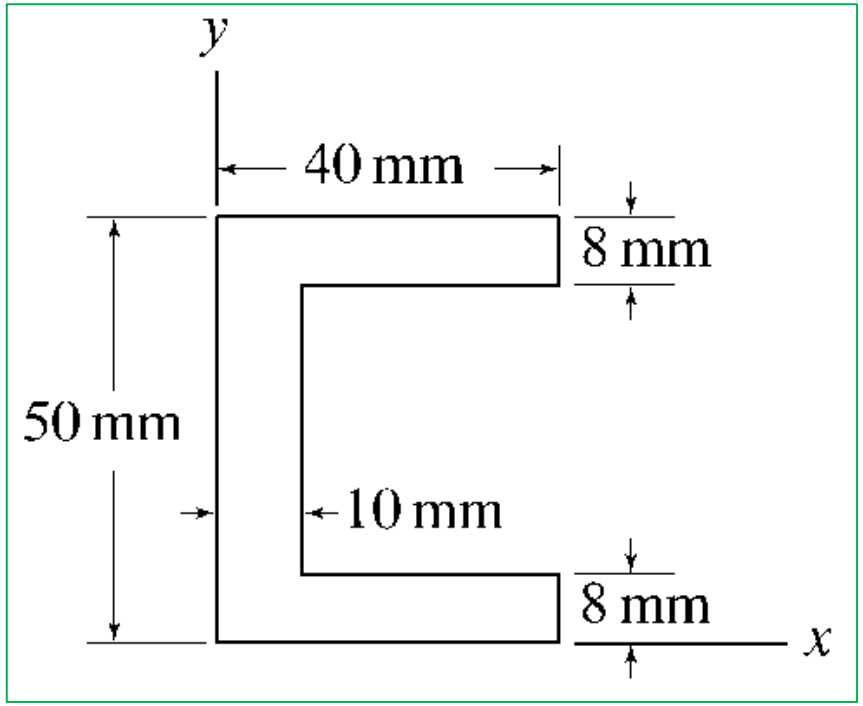
$$\bar{x} = \frac{\sum_{i=1}^{n} \tilde{x}_i A_i}{\sum_{i=1}^{n} A_i} = \frac{\tilde{x}_1 A_1 + \tilde{x}_2 A_2 + \tilde{x}_3 A_3}{A_1 + A_2 + A_3}$$

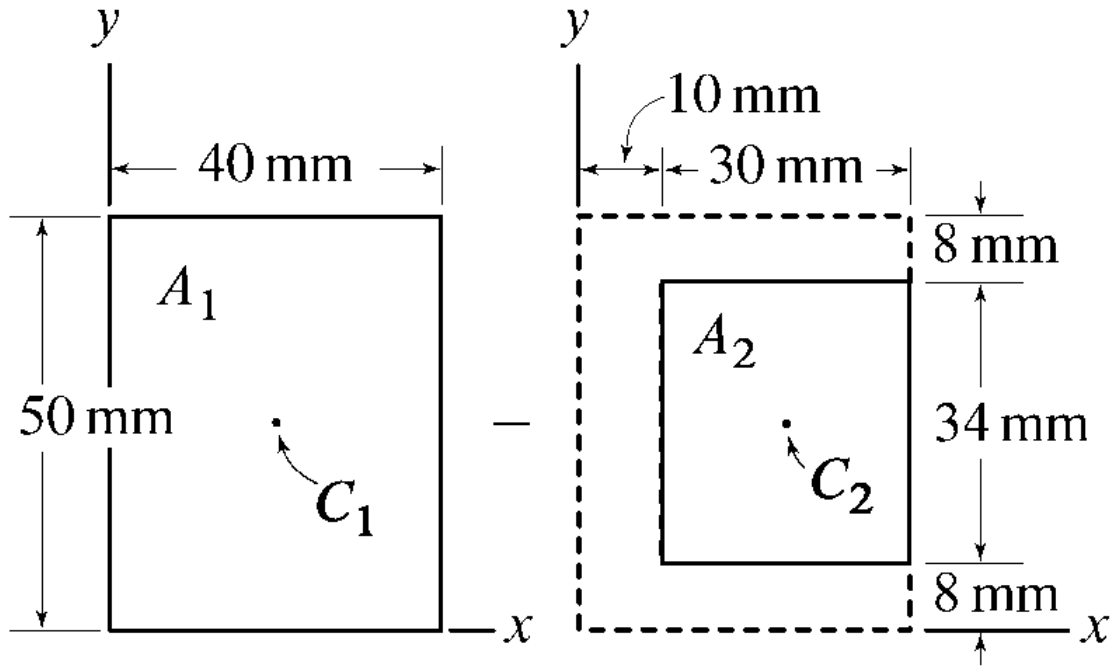
$$\bar{y} = \frac{\sum_{i=1}^{n} \tilde{y}_i A_i}{\sum_{i=1}^{n} A_i} = \frac{\tilde{y}_1 A_1 + \tilde{y}_2 A_2 + \tilde{y}_3 A_3}{A_1 + A_2 + A_3}$$

$$\bar{x} = \frac{(5\ \text{mm})(10\ \text{mm} \times 50\ \text{mm}) + (25\ \text{mm})(8\ \text{mm} \times 30\ \text{mm}) + (25\ \text{mm})(8\ \text{mm} \times 30\ \text{mm})}{(10\ \text{mm} \times 50\ \text{mm}) + (8\ \text{mm} \times 30\ \text{mm}) + (8\ \text{mm} \times 30\ \text{mm})}$$

$$\bar{x} = 14.8\ \text{mm}$$

Example



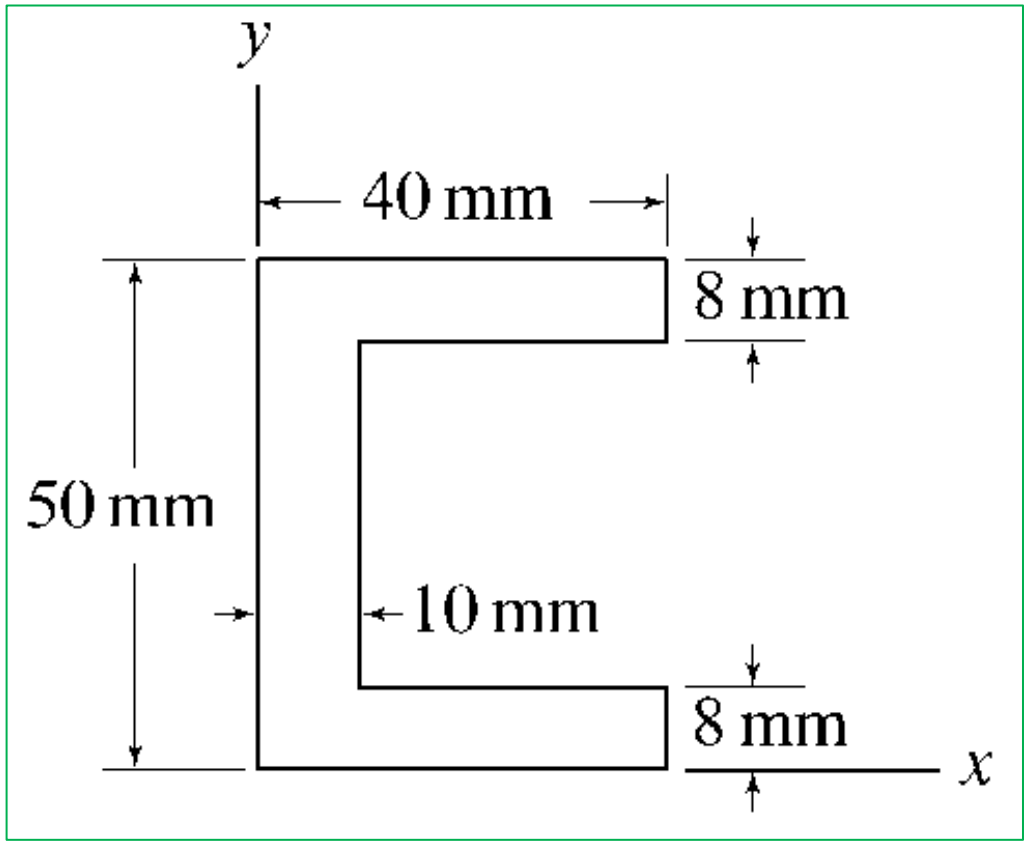


$$\bar{x} = \frac{\tilde{x}_1 A_1 - \tilde{x}_2 A_2}{A_1 - A_2}$$

$$= \frac{(20\ \text{mm})(40\ \text{mm} \times 50\ \text{mm}) - (25\ \text{mm})(30\ \text{mm} \times 34\ \text{mm})}{(40\ \text{mm} \times 50\ \text{mm}) - (30\ \text{mm} \times 34\ \text{mm})}$$

$$= 14.8\ \text{mm}$$

Example

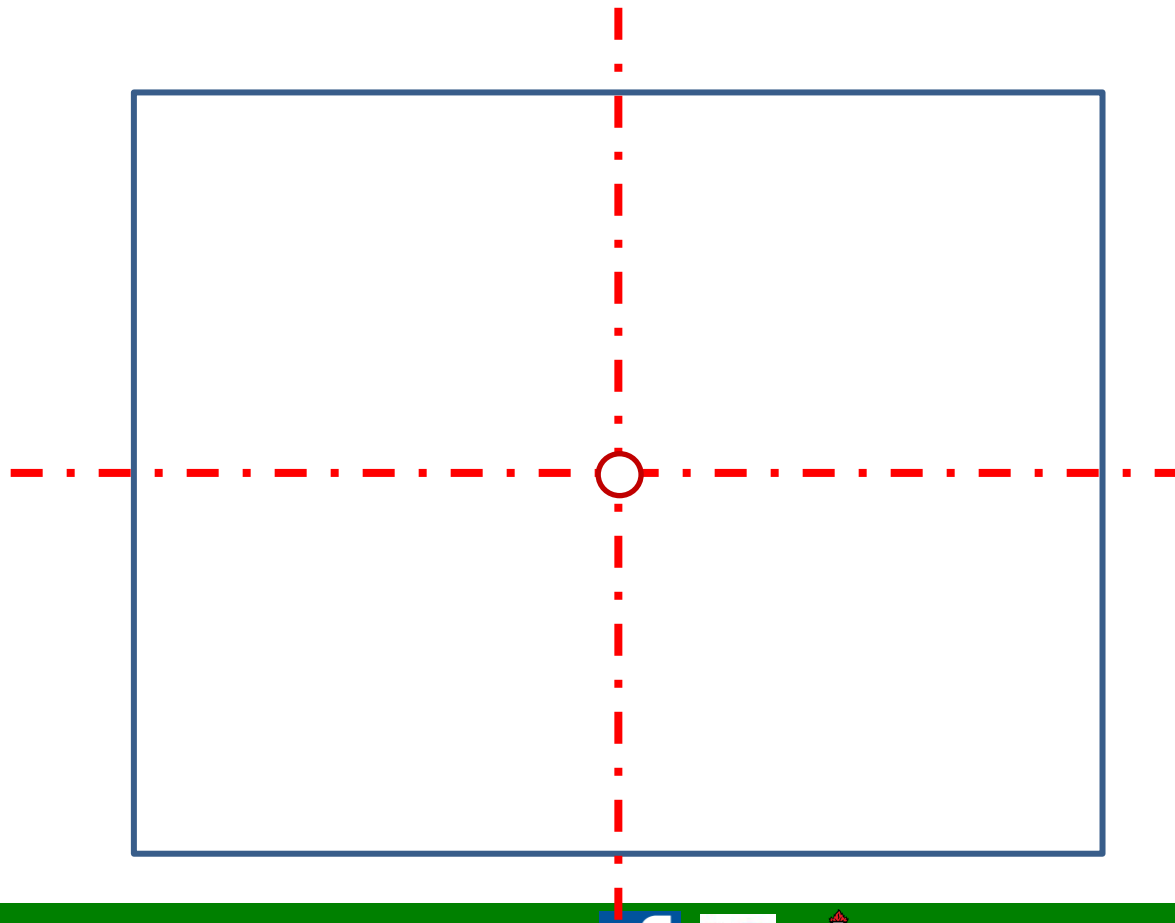


$\bar{y}$ **may be found in a similar manner.**

Centre of Mass of Symmetrical Objects

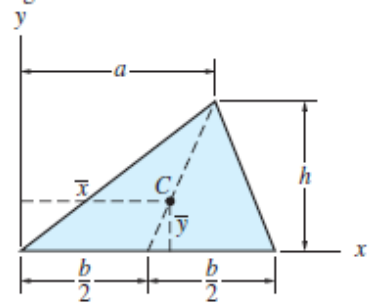
If an object has an axis of symmetry, (i.e. both the object's shape and its mass distributions are symmetric), then the center of mass lies on the axis of symmetry.

If the object has two axes of symmetry, then the center of mass is located at the intersection of the axes of symmetry



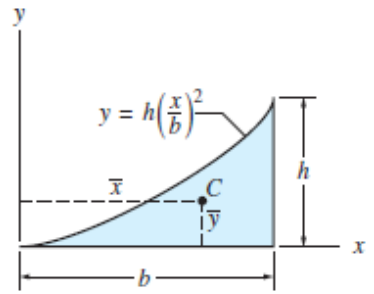
Centroids of some common plane geometries.

Triangle



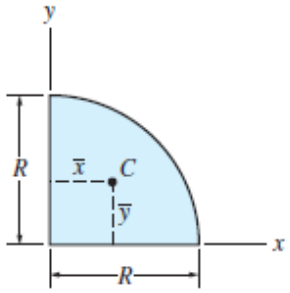
$\bar{x} = \frac{1}{3}(a+b)$ $\bar{y} = \frac{1}{3}h$ $A = \frac{1}{2}bh$

Half parabolic complement



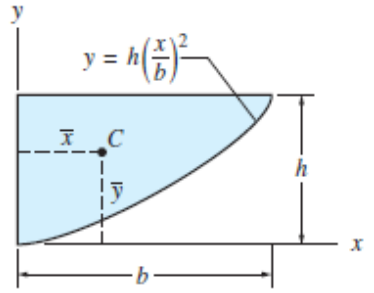
$\bar{x} = \frac{3}{4}b$ $\bar{y} = \frac{3}{10}h$ $A = \frac{1}{3}bh$

Quarter circle



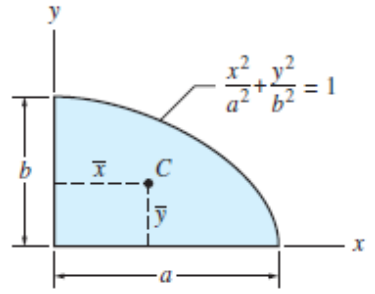
$\bar{x} = \frac{4}{3\pi}R$ $\bar{y} = \frac{4}{3\pi}R$ $A = \frac{\pi}{4}R^2$

Half parabola



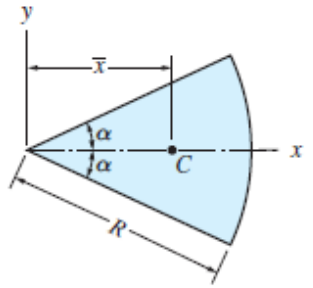
$\bar{x} = \frac{3}{8}b$ $\bar{y} = \frac{3}{5}h$ $A = \frac{2}{3}bh$

Quarter ellipse



$\bar{x} = \frac{4}{3\pi}a$ $\bar{y} = \frac{4}{3\pi}b$ $A = \frac{\pi}{4}ab$

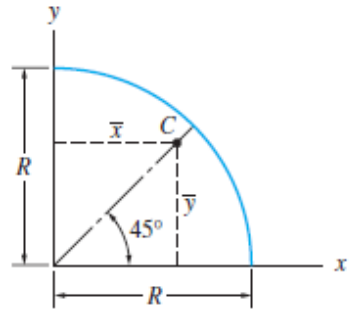
Circular sector



$\bar{x} = \frac{2R\sin\alpha}{3\alpha}$ $A = \alpha R^2$

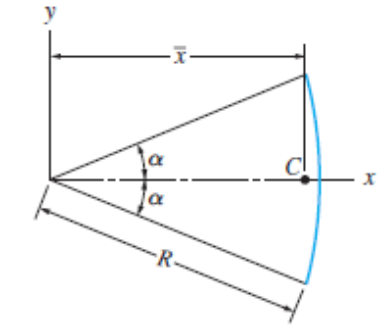
Centroids of plane areas

Quarter circular arc



$\bar{x} = \frac{2}{\pi}R$ $y = \frac{2}{\pi}R$ $L = \frac{\pi}{2}R$

Circular arc



$\bar{x} = \frac{1}{\alpha}R\sin\alpha$ $L = 2\alpha R$

Centroids of plane curves